1) Symmetric Links ($\tau = \tau_{tx} = \tau_{rx},\ p = p_{tx} = p_{rx}$)

Scenario

Symmetric communication link.

Expected Time to Crash Bound

Theorem 1 (Expected Time to Crash Bound)**.**

Safe operation is guaranteed if the expected communication delay does not exceed half of the available reaction time:

$$2\mathbb{E}[t] = \tau_{e2e} + T\frac{1+p}{1-p} \le \Delta t$$

Where

$$\Delta t = 1000 \cdot \frac{\Delta d}{v_{max}} \qquad \mathbb{E}[t] = \tau + T\frac{1+p}{2(1-p)}$$

Proof

The robot must stop before $d_{max} = m + d_{brake}$. Given the threshold S, the distance available for travel at constant velocity v_{max} is $\Delta d = S - d_{max}$. This yields a time budget:

$$\boxed{\Delta t = 1000 \cdot \frac{\Delta d}{v_{max}}\ [ms]}$$

(Note: 1000 is a scaling factor for milliseconds).

We model the delivery of a packet as a sequence of Bernoulli trials. Denote the failure probability as p (packet loss), and $(1-p)$ be the probability of success.

Let $n \sim \mathrm{Geom}(1-p)$ be a discrete random variable representing the number of failed attempts before the first successful delivery. The probability that the first success occurs exactly at the n-th retry is:

$$P(n) = p^n(1-p)$$

The time of arrival t_n for a packet that succeeds after n failures, accounting for a fixed propagation latency τ and a retransmission period T, is defined as:

$$t_n = \tau + nT$$

Using the LOTUS, the expected time of arrival $\mathbb{E}[t_{received}]$ is.

$$\begin{aligned}
\mathbb{E}[t_{received}] &= \sum_{n=0}^{\infty} t_n \cdot P(n) \\
&= \sum_{n=0}^{\infty} (\tau + nT)p^n(1-p) \\
&= \tau(1-p)\sum_{n=0}^{\infty} p^n + T(1-p)\sum_{n=0}^{\infty} np^n \\
&= \tau(1-p)\left(\frac{1}{1-p}\right) + T(1-p)\left(\frac{p}{(1-p)^2}\right)
\end{aligned}$$

Simplifying the terms:

$$\mathbb{E}[t_{received}] = \tau + \frac{pT}{1-p}$$

In a real-time system with a fixed sampling period T, the crossing of the threshold distance S and the controller reception of the signal are asynchronous with the transmission clock. Let $\phi \sim \mathcal{U}(0,T)$ be the random variable representing the phase offset between the event detection and the start of the next transmission slot.

The expected additional delay due to this misalignment is:

$$\mathbb{E}[t_{processing}] = \int_0^T \phi \cdot f(\phi) d\phi = \frac{1}{T}\left[\frac{\phi^2}{2}\right]_0^T = \frac{T}{2}$$

Finally:

$$\boxed{\mathbb{E}[t] = \mathbb{E}[t_{received}] + \mathbb{E}[t_{processing}] = \tau + T\frac{1+p}{2(1-p)}}$$

□

Expected Minimum Distance

Theorem 2 (Expected Minimum Distance)**.**

The expected minimum distance reached from the bot to the obstacle is:

$$\mathbb{E}[d_{\min}] = S - L - d_{brake} - \frac{2\mathbb{E}[t]v_{\max}}{1000}$$

PROOF

Let $X, Y \sim \mathrm{Geom}(1-p)$ be two independent geometric random variables representing the number of failed attempts for the sensor and brake packets, respectively and $\phi \sim \mathcal{U}(0,T)$ the asynchronous processing offset.
The total time to send and receive the signal correctly is:

$$t_{tot} = XT + \tau + YT + \tau + 2\phi = (X+Y)T + \tau_{e2e} + 2\phi$$

Where the expectation yields:

$$\begin{aligned}
\mathbb{E}[t_{tot}] &= (\mathbb{E}[X] + \mathbb{E}[Y])T + \tau_{e2e} + 2\mathbb{E}[\phi] \\
&= 2\frac{p}{1-p}T + \tau_{e2e} + 2\frac{T}{2} \\
&= \frac{1+p}{1-p}T + \tau_{e2e} \\
&= 2\mathbb{E}[t]
\end{aligned}$$

Then the expected time to brake is:

$$\mathbb{E}[d_{\min}] = S - L - d_{brake} - \frac{2\mathbb{E}[t]v_{\max}}{1000}$$

□

Variances

Theorem 3 (Variances)**.**

The reaction time has variance:

$$\mathrm{Var}[t] = T^2 \left(\frac{p}{(1-p)^2} + \frac{1}{12} \right)$$

While the end to end variance is:

$$\mathrm{Var}(t_{e2e}) = T^2 \left(\frac{2p}{(1-p)^2} + \frac{1}{3} \right)$$

Therefore the minimum distance has variance:

$$\mathrm{Var}(d_{min}) = \left(\frac{v_{max}}{1000} \right)^2 T^2 \left(\frac{2p}{(1-p)^2} + \frac{1}{3} \right)$$

PROOF

Proof of reaction time:
The time of arrival is $t_n = \tau + nT$, where $n \sim \mathrm{Geom}(1-p)$. In this formulation, n is the number of failures (starting from 0).

The variance of a geometric distribution n where $P(n) = p^n(1-p)$ is:

$$\mathrm{Var}(n) = \frac{p}{(1-p)^2}$$

Since τ is a constant and T is a scaling factor:

$$\mathrm{Var}(t_{received}) = \mathrm{Var}(\tau + nT) = T^2 \mathrm{Var}(n) = \frac{pT^2}{(1-p)^2}$$

The phase offset ϕ follows a continuous uniform distribution $\mathcal{U}(0, T)$. The variance for any uniform distribution $\mathcal{U}(a, b)$ is $\frac{(b-a)^2}{12}$.

$$\mathrm{Var}(t_{processing}) = \mathrm{Var}(\phi) = \frac{(T-0)^2}{12} = \frac{T^2}{12}$$

Assuming the network packet loss and the clock phase offset are independent:

$$\mathrm{Var}[t] = \mathrm{Var}(t_{received}) + \mathrm{Var}(t_{processing}) = T^2 \left(\frac{p}{(1-p)^2} + \frac{1}{12} \right)$$

Proof of variance of distance:
Since $d_{min} = S - L - d_{brake} - \frac{v_{max}}{1000} \cdot t_{tot}$ has S, L, d_{brake}, and v_{max} who are constants, the variance depends entirely on the total time

$$t_{tot} = (X+Y)T + \tau_{e2e} + 2\phi$$

notice the following:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) = \frac{2p}{(1-p)^2}$$

$$\mathrm{Var}(2\phi) = 4 \cdot \mathrm{Var}(\phi) = 4 \cdot \frac{T^2}{12} = \frac{T^2}{3}$$

then

$$\mathrm{Var}(t_{tot}) = T^2 \mathrm{Var}(X+Y) + \mathrm{Var}(2\phi) = \frac{2pT^2}{(1-p)^2} + \frac{T^2}{3}$$

and finally:

$$\mathrm{Var}(d_{min}) = \left(\frac{v_{max}}{1000}\right)^2 \cdot \mathrm{Var}(t_{tot}) = \left(\frac{v_{max}}{1000}\right)^2 T^2 \left(\frac{2p}{(1-p)^2} + \frac{1}{3}\right)$$

Crash Probability

Theorem 4 (Crash Probability)**.**

Let N_τ be the transmission attempt budget:

$$N_\tau = \left\lfloor \frac{\Delta t - \tau_{e2e}}{T} \right\rfloor$$

Then the probability of a crash ignoring quantization offset is:

$$P_{crash}(p, N_\tau) = p^{N_\tau+1}[1 + (N_\tau + 1)(1-p)]$$

By adding the quantization offset, the probability becomes:

$$P'_{crash}(p, N_\tau) = p^{N_\tau}[1 + N_\tau(1-p)]$$

Proof

Let n be the number of attempts for the sensor packet (X) and m be the number of attempts for the brake command (Y).
The time for a successful transmission is:

$$t_I = (i-1)T + \tau$$

The system avoids a crash if $t_x + t_y \leq \Delta t$. Substituting the expressions:

$$(x + y - 2)T + 2\tau \leq \Delta t \to (x + y) \leq \frac{\Delta t - 2\tau}{T} + 2$$

We define the **total attempt budget** N as:

$$N = \left\lfloor \frac{\Delta t - 2\tau}{T} \right\rfloor + 2$$

A crash occurs if the total required attempts $X + Y > N$

We model X, Y as independent Geometric random variables starting at 1, with failure probability p:

$$P(X = k) = (1-p)p^{k-1}, \quad k \in \{1, 2, \ldots\}$$

Even though Y starts after X succeeds, the *probability* of Y succeeding in its own attempts is independent of X. A crash occurs in two mutually exclusive scenarios:

- Data Fail ($P_{\text{data fail}}$)
- Brake Fail ($P_{\text{brake fail}}$)

Scenario A: Data Fail ($P_{\text{data fail}}$)
This occurs if the sensor data X takes so long that there is no time left for even a single attempt of Y. Since Y requires at least $m = 1$ attempt, X must succeed in $N - 1$ attempts or fewer. If $X \geq N$, the system crashes.

$$P_{\text{data fail}} = P(X > N - 1) = p^{N-1}$$

Scenario B: Brake Fail ($P_{\text{brake fail}}$)
This occurs if X succeeds at attempt $n \in \{1, \ldots, N-1\}$, but the brake command Y fails to succeed within the remaining $N-n$ attempts.

$$\begin{aligned}
P_{\text{brake fail}} &= \sum_{n=1}^{N-1} P(X=n) \cdot P(Y > N-n) \\
&= \sum_{n=1}^{N-1} \left[(1-p)p^{n-1}\right] \cdot \left[p^{N-n}\right] \\
&= \sum_{n=1}^{N-1} (1-p)p^{N-1} \\
&= (N-1)(1-p)p^{N-1}
\end{aligned}$$

The total probability is the sum of both failure modes:

$$P_{\text{crash}}(p, N) = p^{N-1}\left[1 + (N-1)(1-p)\right]$$

Additionally one can include the mean asynchronous processing offset $\mathbb{E}[\phi] = T/2$ as an additive term to the total time on each link:

$$t'_I = (i-1)T + \tau + \frac{T}{2}$$

This allows to find:

$$t'_x + t'_y = (x+y-1)T + \tau \rightarrow N' = \left\lfloor \frac{\Delta t - 2\tau}{T} \right\rfloor + 1$$

For the sake of notation we define

$$N_\tau = \left\lfloor \frac{\Delta t - 2\tau}{T} \right\rfloor \rightarrow \begin{cases} N = N_\tau + 2 \\ N' = N_\tau + 1 \end{cases}$$

and the crash probability becomes

- Case no ϕ:

$$P_{\text{crash}}(p, N_\tau) = p^{N_\tau + 1}\left[1 + (N_\tau + 1)(1-p)\right]$$

- Case ϕ:

$$P'_{crash}(p, N_\tau) = p^{N_\tau}[1 + N_\tau(1-p)]$$

□

S.M.A.R.T. Algorithm

Corollary 5 (S.M.A.R.T. Algorithm)**.**

Let $P_{target} \ll 1$ denote the desired crash probability of a mobile robot under the SMART implementation, then its ideal threshold should be:

$$S = \frac{v_{\max}}{1000}\left(T(N^* + 1) + \tau_{e2e}\right) + L + m + d_{brake}$$

Where N^* is the first $N \in \mathbb{N}^+$ that satisfies $P(N) = p^N(1 + (1-p)N) \leq P_{target}$

PROOF

First notice that $P(N)$ is a monotonically decreasing function. By looking at the difference:

$$P(N) - P(N+1) = p^N (1-p)^2 (N+1) > 0$$

Moreover at the extremes we have

$$P(N=0) = 1 \qquad \lim_{N \to \infty} P(N) = 0$$

Considering N as a continuous variable $N \in \mathbb{R}^+$, since $P(N)$ is a continuous and strictly decreasing function ranging from 1 to 0, by the Intermediate Value Theorem, there exists a unique value $N^* \in \mathbb{R}^+$ such that:

$$P(N^*) = P_{target}$$

For the discrete case required by the algorithm ($N \in \mathbb{N}^+$), the optimal value will be the smallest integer N such that:

$$P(N) \leq P_{target}$$

2) Non Symmetric Links ($\tau_{tx} \neq \tau_{rx}, p_{tx} \neq p_{rx}$)

Expected Time to Crash Bound

Theorem 6 (Expected Time to Crash Bound)**.**

Safe operation is guaranteed if the expected communication delay does not exceed half of the available reaction time:

$$\mathbb{E}[t_{e2e}] = \tau_{e2e} + \frac{T}{2}\left(\frac{1+p_{tx}}{1-p_{tx}} + \frac{1+p_{rx}}{1-p_{rx}}\right) = \tau_{e2e} + T\left(\frac{1-p_{tx}p_{rx}}{1-p_{e2e}}\right) \leq \Delta t$$

Where

$$\Delta t = 1000 \cdot \frac{\Delta d}{v_{max}} \qquad \mathbb{E}[t_i] = \tau_i + T\frac{1+p_i}{2(1-p_i)} \qquad \mathbb{E}[t_{e2e}] = \mathbb{E}[t_{tx}] + \mathbb{E}[t_{rx}]$$

Proof

Same as before just by noticing that $2\mathbb{E}[t] \neq \mathbb{E}[e2e]$ doesn't hold anymore as each link has different parameters.

$$\mathbb{E}[t_i] = \tau_i + T\frac{1+p_i}{2(1-p_i)} \rightarrow \mathbb{E}[t_{tx}] + \mathbb{E}[t_{rx}] \leq \Delta t$$

Moreover with $p_1 = p_2 = p$ we get the result of the symmetric link case. Most notably $\mathbb{E}[t_{e2e}] = 2\mathbb{E}[t]$

□

Expected Minimum Distance

Theorem 7 (Expected Minimum Distance)**.**

The expected minimum distance reached from the bot to the obstacle is:

$$\mathbb{E}[d_{\text{min}}] = S - L - d_{brake} - \frac{\mathbb{E}[t_{e2e}]v_{\text{max}}}{1000}$$

Proof

Direct conclusion from the previous theorem.

□

Variances

Theorem 8 (Variances)**.**

The reaction time on a link has variance:

$$\text{Var}[t_i] = T^2\left(\frac{p_i}{(1-p_i)^2} + \frac{1}{12}\right)$$

While the end to end variance is:

$$\mathrm{Var}[t_{e2e}] = T^2 \left(\frac{p_{tx}}{(1-p_{tx})^2} + \frac{p_{rx}}{(1-p_{rx})^2} + \frac{1}{6} \right)$$

Therefore the minimum distance has variance:

$$\mathrm{Var}(d_{min}) = \left(\frac{v_{max}}{1000} \right)^2 T^2 \left(\frac{p_{tx}}{(1-p_{tx})^2} + \frac{p_{rx}}{(1-p_{rx})^2} + \frac{1}{6} \right)$$

Proof

For each individual link, the variance is calculated based on its specific failure probability p_i and latency τ_i:

$$\mathrm{Var}[t_i] = T^2 \left(\frac{p_i}{(1-p_i)^2} + \frac{1}{12} \right)$$

The total time t_{e2e} is the sum of the uplink delay, the downlink delay, and the processing offsets. Based on your updated model where $\mathbb{E}[t_{e2e}] = \mathbb{E}[t_{tx}] + \mathbb{E}[t_{rx}]$, we assume $t_{e2e} = t_{tx} + t_{rx}$.

$$\begin{aligned}
\mathrm{Var}[t_{e2e}] &= \mathrm{Var}[t_{tx}] + \mathrm{Var}[t_{rx}] \\
&= T^2 \left(\frac{p_{tx}}{(1-p_{tx})^2} + \frac{1}{12} + \frac{p_{rx}}{(1-p_{rx})^2} + \frac{1}{12} \right) \\
&= T^2 \left(\frac{p_{tx}}{(1-p_{tx})^2} + \frac{p_{rx}}{(1-p_{rx})^2} + \frac{1}{6} \right)
\end{aligned}$$

then

$$\mathrm{Var}(d_{min}) = \left(\frac{v_{max}}{1000} \right)^2 T^2 \left(\frac{p_{tx}}{(1-p_{tx})^2} + \frac{p_{rx}}{(1-p_{rx})^2} + \frac{1}{6} \right)$$

Crash Probability

Theorem 9 (Crash Probability)**.**

Let N_τ be the transmission attempt budget:

$$N_\tau = \left\lfloor \frac{\Delta t - \tau_{e2e}}{T} \right\rfloor$$

Then the probability of a crash ignoring quantization offset is (assuming $p_{tx} \neq p_{rx}$):

$$P_{crash}(p_{tx}, p_{rx}, N_\tau) = p_{tx}^{N_\tau+1} + (1-p_{tx}) p_{rx} \frac{p_{rx}^{N_\tau+1} - p_{tx}^{N_\tau+1}}{p_{rx} - p_{tx}}$$

By adding the quantization offset, the probability becomes:

$$P'_{crash}(p_{tx}, p_{rx}, N_\tau) = p_{tx}^{N_\tau} + (1-p_{tx}) p_{rx} \frac{p_{rx}^{N_\tau} - p_{tx}^{N_\tau}}{p_{rx} - p_{tx}}$$

Let n be the number of attempts for the sensor packet (X) and m be the number of attempts for the brake command (Y).
The time for a successful transmission is:

$$t_I = (i-1)T + \tau_i$$

The system avoids a crash if $t_x + t_y \leq \Delta t$. Substituting the expressions:

$$(x+y-2)T+\tau_{e2e} \leq \Delta t \to (x+y) \leq \frac{\Delta t - \tau_{e2e}}{T}+2$$

For the sake of notation we define

$$N_\tau = \left\lfloor \frac{\Delta t - \tau_{e2e}}{T} \right\rfloor \to \begin{cases} N = N_\tau + 2 \\ N' = N_\tau + 1 \end{cases}$$

We model X, Y as independent Geometric random variables starting at 1, with failure probability p_i:

$$P(A_i = k) = (1-p_i)p_i^{k-1}, \quad k \in \{1, 2, \dots\}$$

Even though Y starts after X succeeds, the *probability* of Y succeeding in its own attempts is independent of X. A crash occurs in two mutually exclusive scenarios:

- Data Fail ($P_{\text{data fail}}$)
- Brake Fail ($P_{\text{brake fail}}$)

Scenario A: Data Fail ($P_{\text{data fail}}$)
This occurs if the sensor data (X) uses up so much of the budget that there is no time left for even a single brake attempt ($Y = 1$). Therefore, X must succeed in $N-1$ attempts or fewer. If $X \geq N$, the system crashes immediately.

$$P_{\text{data fail}} = P(X \geq N) = p_{tx}^{N-1}$$

Scenario B: Brake Fail ($P_{\text{brake fail}}$)
This occurs if the sensor data succeeds at some attempt n (where $n < N$), but the brake command fails to succeed within the remaining $N-n$ attempts.

$$\begin{aligned}
P_{\text{brake fail}} &= \sum_{n=1}^{N-1} P(X=n) \cdot P(Y > N-n) \\
&= \sum_{n=1}^{N-1} \left[(1-p_{tx})p_{tx}^{n-1}\right] \cdot \left[p_{rx}^{N-n}\right] \\
&= (1-p_{tx})p_{rx}^{N-1} \sum_{n=1}^{N-1} \frac{p_{tx}^{n-1}}{p_{rx}^{n-1}} \\
&= (1-p_{tx})p_{rx}^{N-1} \sum_{n=1}^{N-1} \left(\frac{p_{tx}}{p_{rx}}\right)^{n-1} \\
&= (1-p_{tx})p_{rx}^{N-1} \left[\frac{1-(p_{tx}/p_{rx})^{N-1}}{1-p_{tx}/p_{rx}}\right] \\
&= (1-p_{tx})p_{rx} \frac{p_{rx}^{N-1} - p_{tx}^{N-1}}{p_{rx} - p_{tx}}
\end{aligned}$$

finally:

$$P_{\text{crash}}(p_{tx}, p_{rx}, N) = p_{tx}^{N-1} + (1-p_{tx})p_{rx} \frac{p_{rx}^{N-1} - p_{tx}^{N-1}}{p_{rx} - p_{tx}}$$

And therefore the two adjusted formulas are correct.

Moreover the symmetric link case holds when substituting at beginning of proof.

□

S.M.A.R.T Algorithm

Corollary 10 (S.M.A.R.T. Algorithm)**.**

Let $P_{target} \ll 1$ denote the desired crash probability of a mobile robot under the SMART implementation, then its ideal threshold should be:

$$S = \frac{v_{\max}}{1000}(T(N^* + 1) + \tau_{e2e}) + L + m + d_{brake}$$

Where N^* is the first $N \in \mathbb{N}^+$ that satisfies $P(N) = p_{tx}^N + (1 - p_{tx})p_{rx}\frac{p_{rx}^N - p_{tx}^N}{p_{rx} - p_{tx}}$

PROOF

To prove that $P(N)$ is strictly decreasing, we examine the ratio or the difference. A more elegant way to see the behavior is to rewrite $P(N)$ as:

$$P(N) = \alpha p_{tx}^N + \beta p_{rx}^N$$

where constants α and β depend on the probabilities. Specifically:

$$P(N) = p_{tx}^N \left(1 - \frac{(1 - p_{tx})p_{rx}}{p_{rx} - p_{tx}}\right) + p_{rx}^N \left(\frac{(1 - p_{tx})p_{rx}}{p_{rx} - p_{tx}}\right)$$

Assuming $0 < p_{tx}, p_{rx} < 1$, $P(N)$ is a linear combination of two decaying exponential functions. Since both p_{tx}^N and p_{rx}^N are strictly decreasing functions of N, their sum (given positive coefficients) is also strictly decreasing.

For the specific case where $p_{tx} \to p_{rx}$, this expression converges to the previous form $p^N(1 + (1 - p)N)$, which we already proved is strictly decreasing.

Moreover at the extremes we have

$$P(N = 0) = 1 \qquad \lim_{N \to \infty} P(N) = 0$$

As before, the optimal value N^* is the smallest integer N such that:

$$P(N) \leq P_{target}$$

□